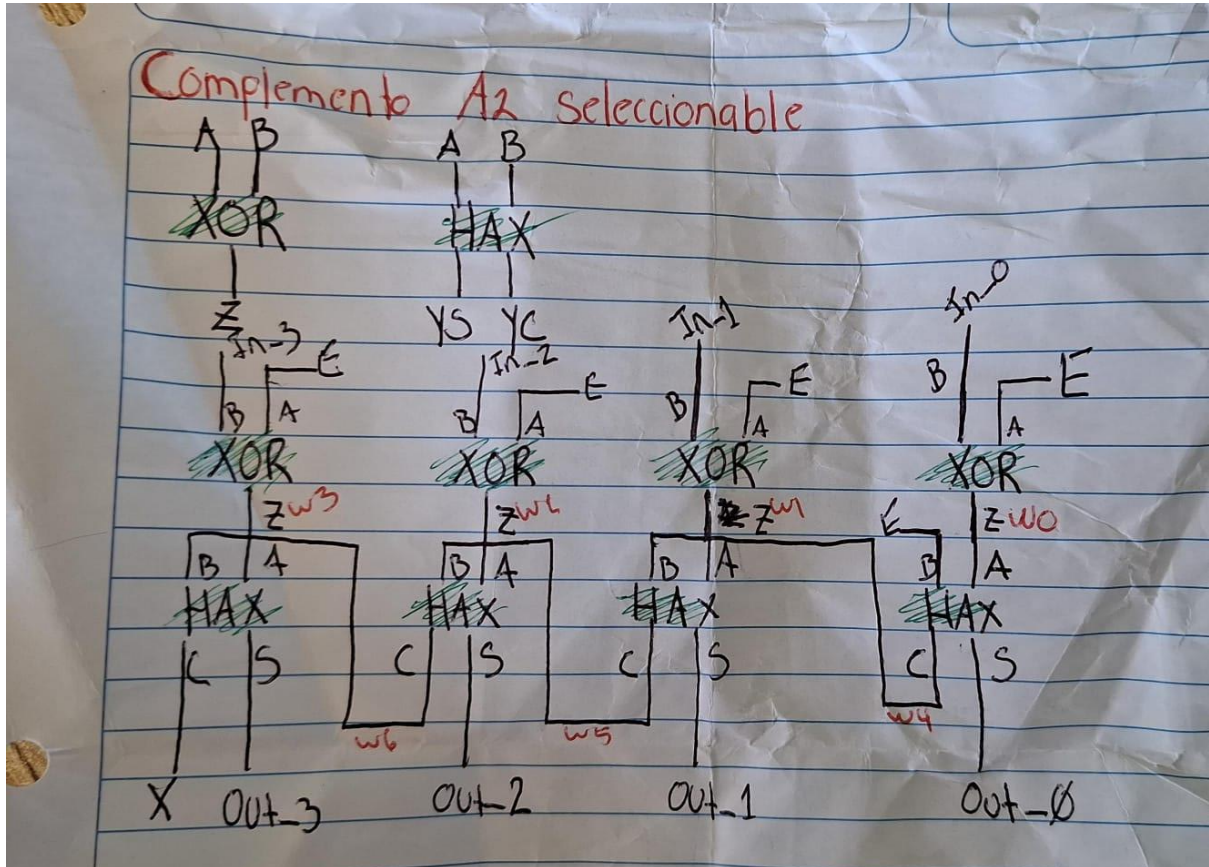


Practice #8: Connect basic cells with Verilog in Cadence

Describe a Verilog circuit in which code uses explicit connections from at least three components that the Genus library has. At the same time the code should have basic description components as the ones used in previous practices.

The circuit we are describing.



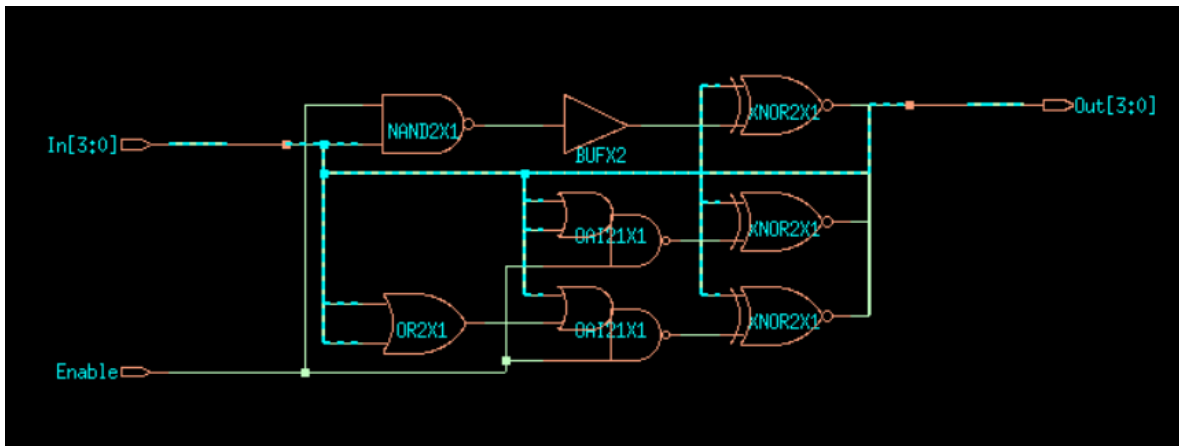
HDL code

```
1 //Verilog HDL for "Verilog", "C_A2_Genus" "functional"
2 module C_A2_Genus (In,Enable,Out);
3
4     input [3:0] In;
5     input Enable;
6     output [3:0] Out;
7     wire w_0, w_1, w_2, w_3, w_4, w_5, w_6;
8
9     XOR_Gates T0 (.A(Enable), .B(In[0]), .Y(w_0));
10    XOR_Gates T1 (.A(Enable), .B(In[1]), .Y(w_1));
11    XOR      T2 (.A(Enable), .B(In[2]), .Y(w_2));
12    XOR      T3 (.A(Enable), .B(In[3]), .Y(w_3));
13
14    HAX1 T4 (.A(w_0), .B(Enable), .YS(Out[0]), .YC(w_4));
15    HAX1 T5 (.A(w_1), .B(w_4), .YS(Out[1]), .YC(w_5));
16    HAX1 T6 (.A(w_2), .B(w_5), .YS(Out[2]), .YC(w_6));
17    HAX1 T7 (.A(w_3), .B(w_6), .YS(Out[3]), .YC());
18
19 endmodule
20
21 module XOR (A,B,Y);
22
23     input A,B;
24     output Y;
25
26     assign Y = A ^ B;
27
28 endmodule
29
30 module XOR_Gates (A,B,Y);
31
32     input A,B;
33     output Y;
34     wire net_0,net_1,net_2,net_3,net_4,net_5;
35
36     INVX1  U0 (.A(A), .Y(net_0));
37     INVX1  U1 (.A(B), .Y(net_1));
38
39     NAND2X1 U2 (.A(B), .B(net_0), .Y(net_2));
40     NAND2X1 U3 (.A(A), .B(net_1), .Y(net_3));
41
42     INVX1  U4 (.A(net_2), .Y(net_4));
43     INVX1  U5 (.A(net_3), .Y(net_5));
44
45     OR2X1  U6 (.A(net_4), .B(net_5), .Y(Y));
46
47 endmodule
```

Testbench code

```
1 //Verilog HDL for "Verilog", "C_A2_Genus_tb" "functional"
2
3 //COMPLEMENTO A 2TestBench
4
5 module C_A2_Genus_tb ();
6
7     reg [3:0] In;
8     reg Enable;
9     wire [3:0] Out;
10
11
12 C_A2_Genus uut(In,Enable,Out);
13
14 initial begin
15
16     $display("A");
17     Enable = 0;
18     In = 4'b0110;
19     #20
20     Enable = 1;
21     #20
22     $display("Test completed");
23
24 end
25
26 endmodule
```

Schematic



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Area report

```
@genus:root: 9> report area
=====
Generated by:      Genus(TM) Synthesis Solution 21.18-s082_1
Generated on:      Oct 16 2024  01:07:35 am
Module:           C_A2_Genus
Technology library: gsc145nm
Operating conditions: typical (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====

Instance  Module  Cell Count  Cell Area  Net Area  Total Area  Wireload
-----
C_A2_Genus      8      26.281    0.000      26.281 <none> (D)
(D) = wireload is default in technology library
@genus:design:C_A2_Genus 10> █
```

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Gates report

```
@genus:design:C_A2_Genus 10> report gates
=====
Generated by:      Genus(TM) Synthesis Solution 21.18-s082_1
Generated on:      Oct 16 2024  01:07:58 am
Module:           C_A2_Genus
Technology library: gscl45nm
Operating conditions: typical (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====

  Gate      Instances    Area      Library
-----
BUFX2             1    2.346    gscl45nm
NAND2X1           1    1.877    gscl45nm
OAI21X1           2    5.632    gscl45nm
OR2X1             1    2.346    gscl45nm
XNOR2X1           3   14.079    gscl45nm
-----
total             8   26.281

  Type          Instances    Area    Area %
-----
buffer              1    2.346     8.9
logic               7   23.934    91.1
physical_cells      0    0.000     0.0
-----
total              8   26.281   100.0

@genus:design:C_A2_Genus 11> █
```

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Power report

```
@genus:design:C_A2_Genus 11> report power
Info    : Joules engine is used. [RPT-16]
Info    : Joules engine is being used for the command report_power.
Info    : ACTP-0001 [ACTPInfo] Activity propagation started for stim#0 netlist
Info    : C_A2_Genus
Info    : ACTP-0009 [ACTPInfo] Activity Propagation Progress Report : 100%
Info    : ACTP-0001 Activity propagation ended for stim#0
Info    : PWRA-0001 [PwrInfo] compute_power effective options
Info    : -mode : vectorless
Info    : -skip_propagation : 1
Info    : -frequency_scaling_factor : 1.0
Info    : -use_clock_freq : stim
Info    : -stim :/stim#0
Info    : -fromGenus : 1
Info    : ACTP-0001 Timing initialization started
Info    : ACTP-0001 Timing initialization ended
Info    : PWRA-0002 [PwrInfo] Skipping activity propagation due to -skip_ap
Info    : option....
Warning: PWRA-0302 [PwrWarn] Frequency scaling is not applicable for vectorless
Warning: PWRA-0304 [PwrWarn] -stim option is not applicable with vectorless mode
Warning: : of power analysis, ignored this option.
Info    : PWRA-0002 Started 'vectorless' power computation.
Info    : PWRA-0002 Finished power computation.
Info    : PWRA-0007 [PwrInfo] Completed successfully.
Info    : Info=6, Warn=2, Error=0, Fatal=0
Instance: /C_A2_Genus
Power Unit: W
PDB Frames: /stim#0/frame#0
-----
Category      Leakage      Internal      Switching      Total      Row%
-----
memory        0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 0.00%
register      0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 0.00%
latch        0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 0.00%
logic        1.36986e-07 9.95312e-07 6.97565e-07 1.82986e-06 100.00%
bbox         0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 0.00%
clock        0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 0.00%
pad          0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 0.00%
pm           0.000000e+00 0.000000e+00 0.000000e+00 0.000000e+00 0.00%
-----
Subtotal     1.36986e-07 9.95312e-07 6.97565e-07 1.82986e-06 100.00%
Percentage   7.49%      54.39%     38.12%     100.00% 100.00%
-----
@genus:design:C_A2_Genus 12> █
```

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Delay report

```
@genus:design:C_A2_Genus 12> report timing -unconstrained
=====
Generated by:      Genus(TM) Synthesis Solution 21.18-s082_1
Generated on:      Oct 16 2024  01:09:18 am
Module:           C_A2_Genus
Operating conditions: typical (balanced_tree)
Wireload mode:    enclosed
Area mode:        timing library
=====

Path 1: UNCONSTRAINED
  Startpoint: (F) In[0]
    Endpoint: (R) Out[3]

      Capture      Launch
  Drv Adjust:++    0        0

      Data Path:-    146

#-----
# Timing Point   Flags   Arc   Edge   Cell       Fanout Load Trans Delay Arrival Instance
#                                     (fF)  (ps)  (ps)  (ps)  Location
#-----
#
  In[0]          -       -     F     (arrival)    4   6.7    0    0      0      (-,-)
  g163/Y         -       B->Y  F     OR2X1        1   2.6   15   48      48      (-,-)
  g159/Y         -       B->Y  R     OAI21X1      1   5.6   59   52     100      (-,-)
  g157/Y         -       B->Y  R     XNOR2X1      1   0.0   28   46     146      (-,-)
  Out[3]         -       -     R     (port)       -   -     -    0     146      (-,-)
#-----

@genus:design:C_A2_Genus 13> █
```

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Code described by Genus

```
// Generated by Cadence Genus(TM) Synthesis Solution 21.18-s082_1
// Generated on: Oct 16 2024 01:11:44 CST (Oct 16 2024 07:11:44 UTC)

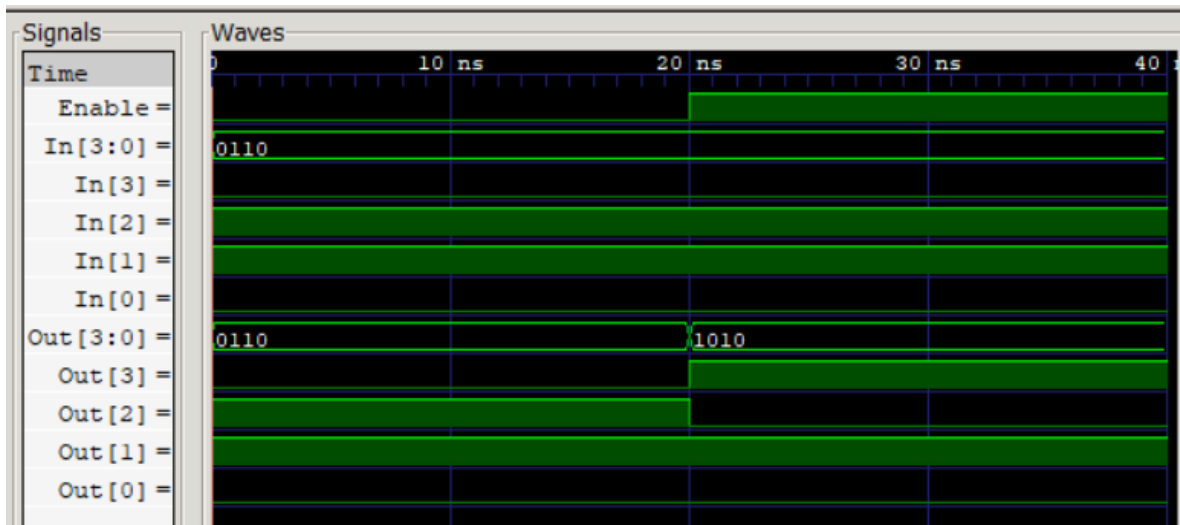
// Verification Directory fv/C_A2_Genus

module C_A2_Genus(In, Enable, Out);
  input [3:0] In;
  input Enable;
  output [3:0] Out;
  wire [3:0] In;
  wire Enable;
  wire [3:0] Out;
  wire n_0, n_1, n_2, n_3, n_13;
  assign Out[0] = In[0];
  XNOR2X1 g157(.A (In[3]), .B (n_3), .Y (Out[3]));
  XNOR2X1 g158(.A (In[2]), .B (n_2), .Y (Out[2]));
  XNOR2X1 g160(.A (In[1]), .B (n_13), .Y (Out[1]));
  OAI21X1 g159(.A (In[2]), .B (n_1), .C (Enable), .Y (n_3));
  OAI21X1 g161(.A (In[0]), .B (In[1]), .C (Enable), .Y (n_2));
  OR2X1 g163(.A (In[1]), .B (In[0]), .Y (n_1));
  NAND2X1 g162(.A (Enable), .B (In[0]), .Y (n_0));
  BUF2X1 drc_bufs(.A (n_0), .Y (n_13));
endmodule
```

It was possible to simulate the circuit?

No, it wasn't possible to simulate. I think that the software couldn't simulate because the code thinks that they are modules from the user but in reality are from part of the libraries and it can't find them so you can't simulate it.

But if we could simulate it we will have something like this:



CODES

//Verilog HDL for "Verilog", "C_A2_Genus" "functional"

```
module C_A2_Genus (In,Enable,Out);

    input [3:0] In;
    input Enable;
    output [3:0]Out;
    wire w_0, w_1, w_2, w_3, w_4, w_5, w_6;

    XOR_Gates T0 (.A(Enable), .B(In[0]), .Y(w_0));
    XOR_Gates T1 (.A(Enable), .B(In[1]), .Y(w_1));
    XOR    T2 (.A(Enable), .B(In[2]), .Y(w_2));
    XOR    T3 (.A(Enable), .B(In[3]), .Y(w_3));

    HAX1 T4 (.A(w_0), .B(Enable), .YS(Out[0]), .YC(w_4));
    HAX1 T5 (.A(w_1), .B(w_4), .YS(Out[1]), .YC(w_5));
    HAX1 T6 (.A(w_2), .B(w_5), .YS(Out[2]), .YC(w_6));
    HAX1 T7 (.A(w_3), .B(w_6), .YS(Out[3]), .YC());

endmodule

module XOR (A,B,Y);

    input A,B;
    output Y;
```

```
assign Y = A ^ B;

endmodule

module XOR_Gates (A,B,Y);

    input A,B;
    output Y;
    wire net_0,net_1,net_2,net_3,net_4,net_5;

    INVX1 U0 (.A(A), .Y(net_0));
    INVX1 U1 (.A(B), .Y(net_1));

    NAND2X1 U2 (.A(B), .B(net_0), .Y(net_2));
    NAND2X1 U3 (.A(A), .B(net_1), .Y(net_3));

    INVX1 U4 (.A(net_2), .Y(net_4));
    INVX1 U5 (.A(net_3), .Y(net_5));

    OR2X1 U6 (.A(net_4), .B(net_5), .Y(Y));

Endmodule
//////////////////////////////////////////////////////////////////////////////////////////////////////////////////
//Verilog HDL for "Verilog", "C_A2_Genus_tb" "functional"

//COMPLEMENTO A 2TestBench

module C_A2_Genus_tb ();

    reg [3:0] In;
    reg Enable;
    wire [3:0] Out;

    C_A2_Genus uut(In,Enable,Out);

initial begin
    $display("A");
    Enable = 0;
    In = 4'b0110;
    #20
    Enable = 1;
    #20
    $display("Test completed");
```

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end

endmodule